

multiplicity-crypto

Test Results Report — Generated from Completed Tests

Phase Mirror Audit
2026-09-20

September 20, 2026

Test Summary

Language	Suite	Passed	Failed
Rust	cargo test	2	0
TypeScript	npm test (vitest run)	17	0
TypeScript	npx tsc --noEmit	1	0
Python	python3 -c import	1	0
Python	pytest / unittest	0	1 (import error)

Total executable assertions: 21 pass, 0 fail. Python full test suite blocked by missing runtime dependencies (numpy, sympy, pytest, pirtm).

1 Rust cargo test

Run from packages/multiplicity-crypto/rust/. Source: rust/src/transcript.rs.

```
Finished 'test' profile [unoptimized + debuginfo] target(s) in 0.02s
Running unittests src/lib.rs (target/debug/deps/multiplicity\_crypto\_rust-8dd48f99c5b1d2)
```

```
running 2 tests
test transcript::tests::test\_different\_messages\_different\_challenges ... ok
test transcript::tests::test\_transcript\_deterministic ... ok
```

```
test result: ok. 2 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out
```

```
Doc-tests multiplicity\_crypto\_rust
running 0 tests
test result: ok. 0 passed; 0 failed; 0 ignored; 0 measured; 0 filtered out
```

2 TypeScript npm test (vitest run)

Run from packages/multiplicity-crypto/ts/. Runner: vitest v4.1.11 (governed by ts/vitest.config.ts with globals: true).

```
> multiplicity-crypto-ts@1.0.0 test
> vitest run
```

```
RUN v4.1.11
```

```
Test Files 5 passed (5)
Tests 17 passed (17)
```

Test files:

- src/__tests__/adr-007-e2e-cryptographic-roundtrip.test.ts (1 test)
- src/__tests__/qkd-roundtrip.test.ts (1 test)
- src/__tests__/feedback.test.ts (6 tests)
- src/__tests__/aead.test.ts (4 tests)
- src/__tests__/frequency.test.ts (5 tests)

```
npx tsc --noEmit: type check passed with zero errors.
```

3 Python python3 -c import multiplicity

```
CAS registry not available, some features disabled
PYTHON\_IMPORT\_OK
```

Package imports cleanly. CAS

```
AVAILABLE : False(expected : pirtm/numpynotininstalled).Allpreviously – unwrappeddead –
moduleimports(agi,topos,cert,utils,moonshine,zenohearthbeat,kernelelemetry)nowgracefullyhandled.3.1
```

```
Blocked. Missing dependencies: numpy, sympy, pytest, pirtm. Declared in py/setup.py
install
```

```
,requiresandextras,require[test].Installwith : pip install -e py/[test].
```

```
ModuleNotFoundError: No module named 'numpy'
```

```
\end{verbatim}
```

```
\section{Dissonance Resolution Status}
```

```
\begin{center}
```

```
\begin{longtable}{p{3.5cm} p{3.5cm} p{2cm} p{4cm}}
```

```
\toprule
```

```
\textbf{Dissonance} & \textbf{Claim (before)} & \textbf{Status} & \textbf{Resolution} \\\
```

```
\midrule
```

```
\endhead
```

```
D-01 & ‘‘QKD Hybrid Encryption’’ & \textbf{Renamed} & Simulated classical hybrid encryption with
```

```
D-02 & ‘‘WASM-backed BN254 Pedersen’’ & \textbf{Documented stub} & SHA-256 stub; BN254 WASM not
```

```
D-03 & Python \_\_\_init\_\_\_ unwrapped imports & \textbf{Fixed} & All wrapped in try/except
```

```
D-04 & vitest not in devDependencies & \textbf{Fixed} & vitest declared; vitest.config.ts with
```

```
D-05 & Contractivity gate untested & \textbf{Fixed} & 6 tests in feedback.test.ts \\\
```

D-06 & LawfulRecursionHash placeholder & \textbf{Documented} & Excluded per ADR-068 F-MC-05 \\
D-07--D-10 & Policy bindings & \textbf{Not addressed} & Governed by companion papers \\
\bottomrule
\end{longtable}
\end{center}

\section{ADR Reference}

\begin{itemize}
\item \textbf{ADR-068}: Phase Mirror Diagnostic --- multiplicity-crypto Label-to-Path Dissonance
\item \textbf{PM-MC-001}: Phase Mirror on multiplicity-crypto and Lawful Recursion v1.0 (\texttt{PM-MC-001.md})
\item \textbf{PHASE_MIRROR_AUDIT}: Engineering mismatch audit (\texttt{PHASE_MIRROR_AUDIT.md})
\item \textbf{TEST_RESULTS}: Test execution record (\texttt{TEST_RESULTS.md})
\end{itemize}

\section*{Generated}

This report was generated from completed test executions on 2026-09-20.
Test commands: \texttt{cargo test}, \texttt{npm test}, \texttt{npm tsc --noEmit}, \texttt{python3 -m unittest}
No tests were mocked, simulated, or fabricated for this report.